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NAAC GRADE A+ ACCREDITED UNIVERSITY

Worksheet 2

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Subject Name: _PL/SQL Lab_
Subject Code: 26CAP-602

1. Aim/Overview of the practical:

Develop a PL/SQL program using an explicit cursor to generate a report of products that are low in stock from an inventory table.

2. Task to be done:

- An explicit cursor `low_stock_cursor` is created to fetch products whose quantity is less than `10`.
- The cursor is opened using `OPEN`.
- Records are retrieved one by one using `FETCH`.
- The loop continues until all matching rows are processed.
- `DBMS_OUTPUT.PUT_LINE` displays the low-stock report.
- Finally, the cursor is closed using `CLOSE`.

3. Software/Tools used:

Oracle SQL Developer / Oracle PL/SQL Environment

4. Practical implementation and steps to be done (with screenshots):

```
-- Step 0:
DROP TABLE Inventory;

-- Step 1: Create Inventory Table
CREATE TABLE Inventory (
Product_ID NUMBER PRIMARY KEY,
Product_Name VARCHAR2(50),
Quantity NUMBER
);

-- Step 2: Insert Sample Records
INSERT INTO Inventory VALUES (101, 'Keyboard', 25);
INSERT INTO Inventory VALUES (102, 'Mouse', 8);
```

```
INSERT INTO Inventory VALUES (103, 'Monitor', 5);
INSERT INTO Inventory VALUES (104, 'Printer', 15);

COMMIT;

-- Step 3: PL/SQL Program using Explicit Cursor
DECLARE
CURSOR low_stock_cursor IS
SELECT Product_ID, Product_Name, Quantity
FROM Inventory
WHERE Quantity < 10;

v_pid   Inventory.Product_ID%TYPE;
v_pname Inventory.Product_Name%TYPE;
v_qty   Inventory.Quantity%TYPE;

BEGIN
OPEN low_stock_cursor;

DBMS_OUTPUT.PUT_LINE('-------------------------------------------');
DBMS_OUTPUT.PUT_LINE('         LOW STOCK PRODUCTS REPORT         ');
DBMS_OUTPUT.PUT_LINE('-------------------------------------------');

LOOP
FETCH low_stock_cursor INTO v_pid, v_pname, v_qty;
EXIT WHEN low_stock_cursor%NOTFOUND;

DBMS_OUTPUT.PUT_LINE(
'Product ID: ' || v_pid ||
' | Product Name: ' || RPAD(v_pname, 10) ||
' | Quantity: ' || v_qty
);
END LOOP;

CLOSE low_stock_cursor;

DBMS_OUTPUT.PUT_LINE('-------------------------------------------');

EXCEPTION
WHEN OTHERS THEN
IF low_stock_cursor%ISOPEN THEN
CLOSE low_stock_cursor;
END IF;
DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
END;
/
```

Result

```
---------------------------------------------
          LOW STOCK PRODUCTS REPORT
---------------------------------------------
Product ID: 102 | Product Name: Mouse      | Quantity: 8
Product ID: 103 | Product Name: Monitor    | Quantity: 5
---------------------------------------------


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.014
```

5. Learning outcomes (What I have learnt):

- Learned how to create an inventory table and insert sample records.
- Understood how to declare an explicit cursor with a `WHERE` condition.
- Used `%TYPE` to anchor PL/SQL variables directly to table column datatypes.
- Learned how to open a cursor and allocate system resources using `OPEN`.
- Understood how to fetch rows one by one into variables using `FETCH`.
- Applied the `LOOP` and `EXIT WHEN %NOTFOUND` statements to control row iteration.
- Used `DBMS_OUTPUT.PUT_LINE()` to format and print the low-stock report.
- Learned the importance of closing the cursor with `CLOSE` to free memory.
- Implemented an `EXCEPTION` block to catch runtime errors safely.
- Understood how to use `%ISOPEN` inside exception handling to prevent cursor leaks.